ECON 6067

Minimum Wages and Employment: A Replication and Extension of Card and Krueger (1994)

Empirical Report

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Abstract

This report replicates and extends Card and Krueger's (1994) difference-in-differences study of minimum wage effects. Exploiting the 1992 minimum wage hike in New Jersey (from $4.25 to $5.05 per hour) with eastern Pennsylvania as a control group, we use survey data from 410 fast-food restaurants to reconstruct the original sample, variables, and empirical design. We replicate the core finding: the wage hike did not reduce fast-food employment; instead, employment rose slightly in New Jersey relative to Pennsylvania. The result is robust across multiple specification checks and a placebo test. We further test for heterogeneity by store ownership using a triple-difference framework. Point estimates show a larger positive effect for franchised stores, consistent with incentive theory, but the difference is statistically insignificant.

1. Research Question and Motivation

The employment impact of minimum wage policies is one of the most longstanding and contested debates in labor economics. Conventional neoclassical theory predicts that a binding minimum wage raises labor costs and reduces employment in competitive markets (Stigler, 1946). By contrast, theories of monopsonistic labor markets suggest that moderate minimum wage increases can raise employment by constraining firms' wage-setting power.

Prior to Card and Krueger (1994), most empirical work relied on aggregate time-series data and found small negative employment effects for teenagers. These studies were limited, however, by potential endogeneity and confounding macroeconomic shocks. Card and Krueger advanced the literature by exploiting a natural experiment: the April 1992 increase in New Jersey's state minimum wage from $4.25 to $5.05 per hour, with neighboring eastern Pennsylvania—where the minimum wage remained unchanged—as a control group. Using surveys of fast-food restaurants before and after the policy change, they implemented a difference-in-differences design to credibly identify the causal effect of the policy.

The original study found no evidence of employment declines in New Jersey relative to Pennsylvania; if anything, employment rose modestly in the treatment state. This finding challenged prevailing consensus and spurred a large body of subsequent research.

This project has two primary goals. First, we replicate the core empirical results of Card and Krueger (1994) using the original survey data, assessing the robustness and replicability of their well-known finding. Second, we extend the analysis by examining whether employment effects differ systematically across company-owned

and franchised restaurants, to test whether organizational incentives shape the impact of minimum wage policy.

2. Data Sources, Sample Construction, and Variable Definitions

2.1 Data Sources

The analysis draws on store-level survey data from the original Card and Krueger (1994) study, provided as part of the course project datasets. The file public.csv contains 410 fast-food restaurant observations, with variables measured across two survey waves. Wave 1 (pre-policy) was fielded in February–March 1992, roughly one month before the minimum wage increase took effect. Wave 2 (post-policy) followed in November–December 1992, approximately seven to eight months after policy implementation. The sample spans four major fast-food chains—Burger King, KFC, Roy Rogers, and Wendy's—across New Jersey (NJ) and eastern Pennsylvania (PA), and includes information on employment, starting wages, product prices, store characteristics, and operational details.

2.2 Sample Construction

The original sample frame comprises 473 stores with working telephone numbers. Of these, 410 completed the first-wave interview, corresponding to an overall response rate of 86.7 percent. All 410 first-wave respondents were followed up for the second survey wave. Among second-wave outcomes, 371 stores completed phone interviews, and an additional 28 stores completed in-person interviews after initially refusing phone participation. Six stores were permanently closed by the time of wave 2, and another 4 were temporarily closed due to renovation, highway construction, or fire.

For the core employment analysis, we follow the original study's protocol: permanent closures are retained in the sample with wave-2 employment set to zero, while temporary closures are coded as missing. The main balanced sample used for regression analysis includes 357 stores with complete employment and wage data for both waves.

2.3 Variable Definitions

Dependent Variables

The primary outcome variable is full-time equivalent (FTE) employment, calculated as:

$$\text{FTE} = \text{Full-time workers} + \text{Managers/ assistant managers} + 0.5 \times \text{Part-time workers}$$

This construction follows the standard convention of weighting part-time workers at 50 percent of full-time workers to reflect typical hours differences. The second key dependent variable is full meal price, defined as the combined price of a medium soda, small fries, and an entree (all including tax), used to analyze price pass-through of the wage increase.

Core Policy Variables

We construct two policy variables for the difference-in-differences analysis. First, NJ is a binary indicator equal to 1 for stores in New Jersey (the treatment state) and 0 for stores in Pennsylvania (the control state). Second, GAP is a continuous measure of policy shock intensity, defined as:

$$GAP_i = \begin{cases} 0 & if\ store\ i\ is in Pennsylvania \\ 0 & if store\ i\ is\ in\ NJ\ and\ initial\ wage \geq \$5.05 \\ \frac{5.05 - WAGE_{i,1}}{WAGE_{i,1}} & if\ store\ i\ is\ in\ NJ\ and\ initial\ wage < \$5.05 \end{cases}$$

This variable captures the percentage wage increase required for compliance with the new minimum, and takes a value of zero for stores already paying at or above the $5.05 threshold.

Control Variables

The set of control variables includes chain indicators—binary variables for Burger King, KFC, Roy Rogers, and Wendy's, with Burger King as the reference group. We also include a company-owned indicator equal to 1 for company-owned stores and 0 for franchised outlets, and region indicators for sub-regions within New Jersey (North, Central, South) and Pennsylvania (Philadelphia suburbs, Easton area). Finally, we control for average daily operating hours on weekdays.

2.4 Data Cleaning Rules

We follow the original paper's protocols for handling missing values and edge cases. For employment outcomes, permanently closed stores (STATUS2 = 3) are retained in the sample with wave-2 FTE set to zero. Temporarily closed stores (STATUS2 = 2, 4, 5), including closures due to renovation, highway construction, and fire, are coded as missing for wave-2 employment. All monetary variables remain in nominal dollars as in the original study, and observations with missing values for core variables are excluded from the relevant analyses.

3. Empirical Design

3.1 Baseline Difference-in-Differences (DID)

Our core empirical strategy is a standard difference-in-differences (DID) design, which compares employment changes in New Jersey (the treatment group) to changes in Pennsylvania (the control group) before and after the minimum wage increase.

The key identifying assumption is that employment trends would have evolved in parallel across the two states in the absence of the policy change. The geographic proximity of the two regions, their similar underlying economic conditions, and their shared seasonal employment patterns all support the plausibility of this parallel trends assumption.

We implement the DID design in two complementary specifications. The first is a state-level binary DID model, specified as:

$$FTE_{ist} = \alpha + \beta \cdot (NJ_s \times Post_t) + \delta \cdot NJ_s + \tau \cdot Post_t + \gamma \cdot X_i + \varepsilon_{ist}$$

where subscript s indexes state, t indexes survey wave, and X_i is a vector of store-level control variables. The coefficient β captures the DID estimate of the minimum wage's causal effect on employment.

The second specification uses a continuous wage gap measure to exploit variation in policy intensity within New Jersey. This first-difference framework is specified as:

$$\Delta FTE_i = \alpha + \beta \cdot GAP_i + \gamma \cdot X_i + \varepsilon_i$$

where $\Delta FTE_i = FTE_{i,2} - FTE_{i,1}$ is the change in employment at store i between the two waves. This specification compares stores within New Jersey that faced larger required wage increases to those that faced smaller shocks or no shock at all, as well as to stores in Pennsylvania where the policy did not apply.

3.2 Triple-Difference Extension

For the independent extension, we extend the baseline DID framework by incorporating a third dimension of heterogeneity: store ownership type. The triple-difference (DDD) model tests whether the employment effect of the minimum wage differs systematically between company-owned and franchised stores. The full specification is:

$$\begin{aligned} FTE_{ist} = \alpha \quad & + \beta_1 NJ_s + \beta_2 Post_t + \beta_3 CoOwned_i \\ & + \beta_4 (NJ_s \times Post_t) + \beta_5 (NJ_s \times CoOwned_i) + \beta_6 (Post_t \times CoOwned_i) \\ & + \beta_7 (NJ_s \times Post_t \times CoOwned_i) + \gamma X_i + \varepsilon_{ist} \end{aligned}$$

The coefficient (β7) is the key parameter of interest, as it measures the differential DID effect for company-owned stores relative to franchised outlets. A negative value of (β7) would indicate smaller employment effects for company-owned stores, consistent with the theoretical prediction that weaker performance incentives and limited operational flexibility reduce efficiency adjustments in response to wage increases.

All regression specifications report heteroskedasticity-robust standard errors.

4. Core Replication Results and Comparison with the Original Paper

4.1 Descriptive Statistics

Table 2 reports means of key variables for New Jersey and Pennsylvania across both survey waves. These statistics first assess pre-policy balance across the treatment and control groups, and then provide descriptive evidence on how wages and employment evolved following the minimum wage increase.

Table2.Means of Key Variables by State and Survey Wave

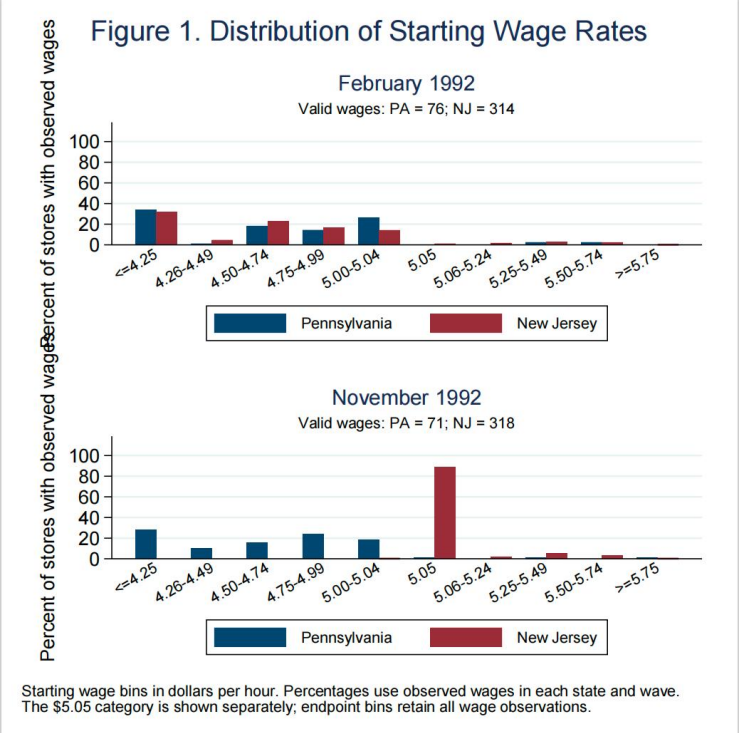
	Stores in:		
Variable	NJ	PA	t (NJ-PA)
1. *Distribution of store types(%)*			
a. BurgerKing	41.09	44.30	-0.5
b. KFC	20.54	15.19	1.2
c. RoyRogers	24.77	21.52	0.6
d. Wendy's	13.60	18.99	-1.1
e. Company-owned	34.14	35.44	-0.2
2. *Means in wave1*			
a. FTE employment	20.44	23.33	-2.0
	(0.51)	(1.35)	
b. Percentage full-time employees	32.85	35.04	-0.7
	(1.33)	(2.73)	
c. Starting wage	4.61	4.63	-0.4
	(0.02)	(0.04)	
d. Wage=$4.25 (%)	30.51	32.91	-0.4
	(2.53)	(5.32)	
e. Price of full meal	3.35	3.04	4.0
	(0.04)	(0.07)	
f. Hours open (weekday)	14.42	14.53	-0.3
	(0.15)	(0.33)	
g. Recruiting bonus	23.56	29.11	-1.0
	(2.34)	(5.14)	
3. *Means in wave2*			
a. FTE employment	21.03	21.17	-0.1
	(0.52)	(0.94)	
b. Percentage Full-time employees	35.87	30.38	1.8
	(1.38)	(2.81)	
c. Starting wage	5.08	4.62	10.8
	(0.01)	(0.04)	
d. Wage=$4.25 (%)	0.00	25.32	--

	(0.00)	(4.92)	
e. Wage=$5.05 (%)	85.50	1.27	36.4
	(1.94)	(1.27)	
f. Price of full meal	3.41	3.03	5.1
	(0.04)	(0.07)	
g. Hours open (weekday)	14.42	14.65	-0.6
	(0.15)	(0.33)	
h. Recruiting bonus	20.32	23.38	-0.6
	(2.27)	(4.85)	

Notes. Standard errors in parentheses. Means use all non-missing observations for each state-wave cell. FTE includes managers plus 0.5 times part-time workers. Permanent closures have wave-2 FTE set to zero; temporary closures are coded as missing.

Before the policy took effect (wave 1), store characteristics were broadly balanced across states. Starting wages were nearly identical: $4.61 in New Jersey and $4.63 in Pennsylvania, a statistically insignificant difference of $0.02. The two states differed more noticeably in store size and price levels: Pennsylvania stores were larger on average, and full meal prices were significantly higher in New Jersey. These level differences motivate the inclusion of store-level controls in later regression analysis.

By wave 2, the minimum wage had produced a stark divergence in wages. Average starting wages rose by $0.47 to $5.08 in New Jersey, while remaining essentially flat at $4.62 in Pennsylvania. Most strikingly, 85.5 percent of New Jersey stores paid exactly $5.05 per hour, compared to only 1.3 percent of Pennsylvania stores. To more clearly visualize how the policy reshaped the full wage structure, Figure 1 plots the distribution of starting wages across states and waves.



Starting wage bins in dollars per hour. Percentages use observed wages in each state and wave. The $5.05 category is shown separately; endpoint bins retain all wage observations.

Notes. Top panel: February 1992 (pre-policy). Bottom panel: November 1992 (post-policy). Dark bars = New Jersey; light bars = Pennsylvania. Percentages calculated relative to all observed wages in each state and wave. Wage bins in dollars per hour. Valid wage observations: PA = 76, NJ = 314 (wave 1); PA = 71, NJ = 318 (wave 2).

In February 1992, the wage distributions in the two states were closely aligned. Both were heavily concentrated between $4.25 and $4.75, and the modal wage in both states was the federal minimum of $4.25. The near-identical pre-policy distributions further support the comparability of the treatment and control groups.

By November 1992, the New Jersey distribution had shifted almost entirely to the new $5.05 minimum, forming a sharp spike at that value. The Pennsylvania distribution, by contrast, remained largely unchanged, still with a large mass at $4.25. This dramatic visual divergence confirms that the minimum wage mandate had a strong, concentrated first-stage effect on wages in the treatment state, providing the necessary foundation for the causal employment analysis that follows.

4.2 DID Mean Estimates

Table 3 presents formal DID mean estimates of employment effects under alternative sample constructions and comparison groups. The left panel reports cross-state comparisons between New Jersey and Pennsylvania, while the right panel reports within-New Jersey comparisons across stores grouped by their initial wage level.

Table 3. Average Employment per Store Before and After the Minimum Wage Increase

Variable	PA	NJ	NJ-PA	Wage $4.25	Wage $4.26-4.99	Wage >=$5.00	Low-high	Mid-high
FTE before, all obs.	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	-2.69 (1.37)	-2.17 (1.41)
FTE after, all obs.	21.17 (0.94)	21.03 (0.52)	-0.14 (1.08)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.66 (1.44)	0.74 (1.27)
Change in mean FTE	-2.17 (1.22)	0.59 (0.47)	2.75 (1.31)	1.32 (0.81)	0.87 (0.67)	-2.04 (0.98)	3.36 (1.27)	2.91 (1.19)
Change, balanced sample	-2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.20 (0.82)	0.71 (0.69)	-2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
Change, temp. closures = 0	-2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	-2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

Notes. Standard errors in parentheses. FTE includes managers plus 0.5 times part-time workers. Permanent closures are set to zero; temporary closures are coded as missing except in the final row.

The baseline DID estimate using all available observations is +2.75 FTE, with a standard error of 1.31, statistically significant at conventional 5 percent levels. The estimate is virtually identical in the balanced sample of stores with complete data in

both waves, at +2.75 FTE. When temporary closures are also coded as zero employment, the estimate falls slightly to +2.51 but remains positive and economically meaningful.

The right panel of Table 3 reports within-state comparisons across stores grouped by their initial wage level, which provides an alternative test that nets out state-level trends. Stores with the lowest initial wages ($4.25)—the group most exposed to the policy—saw employment growth of 1.32 FTE. By contrast, stores with initial wages at or above $5.00, which were largely unaffected by the mandate, saw employment decline of 2.04 FTE. The difference between low-wage and high-wage stores is +3.36 FTE, similar in magnitude to the cross-state DID estimate. This within-state evidence supports the interpretation that the employment increase is driven by the minimum wage shock rather than confounding state-level macroeconomic trends.

4.3 Reduced-Form Regression Results

Table 4 reports ordinary least squares (OLS) regression estimates of the employment effect, with controls added progressively across columns. The first two columns use the binary New Jersey treatment indicator, while columns (iii) through (v) use the continuous wage gap measure.

Table 4. Reduced-Form Models for Change in Employment

Variable	(i)	(ii)	(iii)	(iv)	(v)
New Jersey dummy	2.33 (1.19)	2.30 (1.20)	--	--	--
Initial wage gap	--	--	15.65 (6.08)	14.92 (6.21)	11.98 (7.42)
Chain and ownership controls	no	yes	no	yes	yes
Region controls	no	no	no	no	yes
Standard error of regression	8.79	8.78	8.76	8.76	8.75
Joint control F-test p-value	--	0.34	--	0.44	0.40

Notes. Heteroskedasticity-robust standard errors in parentheses. All models include a constant term. Sample: 357 stores. Dependent variable: change in FTE employment between wave 1 and wave 2. Omitted chain category: Burger King. Omitted region category: North New Jersey.

Column (i) presents the simple DID estimate with no additional controls. The New Jersey coefficient is 2.33 with a robust standard error of 1.19, positive and statistically significant at the 5 percent level. Adding chain and ownership controls in column (ii) leaves the estimate nearly unchanged at 2.30, and the joint F-test for the control variables has a p-value of 0.34. This indicates that store characteristics are well balanced across the two states, so omitted variable bias from these observables is limited.

Columns (iii) through (v) use the continuous wage gap measure to exploit variation in policy intensity. Without controls, the GAP coefficient is 15.65 with a standard error

of 6.08. This implies that a store requiring a 10 percent wage increase to comply with the new minimum experienced employment growth of roughly 1.56 workers. Adding chain and ownership controls reduces the coefficient slightly to 14.92. With full region controls added in column (v), the coefficient falls to 11.98 and loses statistical significance at conventional levels, but remains positive in sign. This attenuation is expected, as region fixed effects absorb some of the cross-sectional variation that identifies the policy effect.

Across all five specifications, the estimated employment effect is consistently positive. This pattern contradicts the standard neoclassical prediction that minimum wage increases reduce employment.

4.4 Robustness Checks

Table 5 presents a battery of specification tests to assess the robustness of the baseline employment result, covering alternative variable definitions, sample restrictions, estimation approaches, and a placebo falsification test.

Table 5. Specification Tests of Employment Models

Specification	Level: NJ	Level: GAP	Proportion: NJ	Proportion: GAP
1. Base specification	2.30 (1.20)	14.92 (6.21)	0.04 (0.06)	0.25 (0.30)
2. Temporary closures = 0	2.03 (1.22)	14.42 (6.31)	0.02 (0.06)	0.26 (0.31)
3. Exclude managers	2.34 (1.17)	14.69 (6.05)	0.04 (0.07)	0.27 (0.37)
4. Part-time weight 0.4	2.34 (1.20)	15.23 (6.23)	0.05 (0.06)	0.32 (0.34)
5. Part-time weight 0.6	2.27 (1.21)	14.60 (6.26)	0.04 (0.05)	0.20 (0.28)
6. Exclude NJ shore stores	2.59 (1.20)	16.88 (6.37)	0.05 (0.06)	0.32 (0.31)
7. Interview timing controls	2.30 (1.20)	16.45 (6.20)	0.04 (0.06)	0.33 (0.30)
8. Exclude wave-1 calls greater than two	2.42 (1.29)	14.08 (7.11)	0.04 (0.06)	0.26 (0.33)
9. Weight proportional change by initial employment	--	--	0.10 (0.05)	0.69 (0.28)
12. Pennsylvania placebo wage gap	--	-0.30 (22.00)	--	-0.66 (0.80)

Notes. All models include chain and ownership controls and a constant term. Robust standard errors in parentheses. Proportional change specifications use initial employment as the denominator.

Rows 2 through 8 evaluate alternative variable constructions and sample restrictions. These include coding temporary closures as zero employment, excluding managers from the FTE calculation, using alternative weights for part-time workers, dropping stores in New Jersey's shore region, controlling for interview timing, and excluding stores that required more than two call-backs in the first wave. Across all seven specifications, both the New Jersey dummy and the wage gap coefficient remain

positive and similar in magnitude to the baseline estimate. None of the alternative specifications yields a significantly negative employment effect.

Row 9 reports results for proportional employment changes weighted by initial store size. These estimates also show positive effects of the minimum wage, confirming that the baseline result is not an artifact of measuring employment in absolute levels.

Row 12 presents results from a placebo test using only Pennsylvania stores. We construct a hypothetical wage gap for Pennsylvania stores as if they had been subject to the same minimum wage increase, and regress employment changes on this placebo measure. The placebo coefficient is −0.30 with a very large standard error of 22.00, and is statistically indistinguishable from zero. This falsification test supports the causal interpretation of our baseline results: there is no systematic relationship between the wage gap measure and employment growth where there was no actual minimum wage shock.

4.5 Additional Outcomes and Price Effects

Beyond employment, we examine effects on other operational margins to explore potential channels of adjustment. Table 6 reports results for a range of outcome measures, including staffing composition, operating hours, employee amenities, and wage profile characteristics.

Table 6. Effects of the Minimum Wage Increase on Other Operational Outcomes

Outcome measure	NJ mean	PA mean	NJ-PA	NJ dummy	Wage gap	Wage gap + regions
Store Characteristics:						
1. Fraction full-time workers (percentage)	2.65 (1.71)	-4.65 (3.80)	7.30 (4.17)	7.30 (3.95)	44.89 (21.05)	38.73 (25.28)
2. Number of hours open per weekday	-0.00 (0.06)	0.11 (0.08)	-0.11 (0.10)	-0.11 (0.12)	-0.30 (0.67)	-0.05 (0.82)
3.Number of cash registers	-0.04 (0.04)	0.13 (0.10)	-0.17 (0.11)	-0.18 (0.10)	-0.30 (0.55)	0.41 (0.67)
4.Number of cash registers open at 11:00 A.M.	-0.04 (0.05)	-0.22 (0.09)	0.18 (0.10)	0.18 (0.12)	0.12 (0.65)	-0.69 (0.79)
Employee Meal Programs:						
5. Low-price meal program (percentage)	-4.67 (2.65)	-1.28 (3.87)	-3.39 (4.69)	-2.13 (5.64)	-22.88 (30.02)	-20.11 (36.44)
6. Free meal program (percentage)	8.41 (2.18)	6.41 (3.33)	2.00 (3.98)	0.49 (4.50)	28.83 (24.38)	36.53 (29.56)
7. Combination of low-price and free meals (percentage)	-4.05 (1.98)	-5.13 (3.11)	1.08 (3.69)	1.20 (4.32)	-14.67 (23.19)	-26.53 (28.06)
Wage Profile:						
8. Time to first raise (weeks)	3.77 (0.89)	1.26 (1.97)	2.51 (2.16)	2.21 (2.03)	0.40 (11.07)	-13.23 (13.42)
9. Usual amount of first raise (cents)	-0.98 (0.93)	-1.75 (1.65)	0.77 (1.89)	0.55 (2.09)	1.12 (11.12)	-1.54 (13.49)
10. Slope of wage profile (percent per week)	-0.10 (0.04)	-0.11 (0.09)	0.00 (0.09)	0.01 (0.09)	-0.08 (0.47)	-0.08 (0.57)

Notes. Standard errors in parentheses. All regression models include chain dummies and a company ownership control.

Overall, there is little evidence of systematic adjustment along non-wage margins. The New Jersey dummy is statistically insignificant for operating hours, number of

cash registers, employee meal programs, and wage profile characteristics such as time to first raise and the slope of the wage-tenure profile. The only marginally significant effect is a small increase in the fraction of full-time workers in New Jersey relative to Pennsylvania. This suggests that stores did not offset higher wage costs by cutting non-wage benefits or scaling back operations.

We next examine the extent to which the minimum wage increase was passed through to product prices. Table 7 reports reduced-form regression results for the change in full meal prices, using the same progressive control structure as the employment regressions.

Table 7. Reduced-Form Models for Change in Full Meal Price

Variable	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	0.033 (0.014)	0.037 (0.014)	--	--	--
2. Initial wage gap	--	--	0.078 (0.075)	0.147 (0.074)	0.066 (0.089)
3. Chain and ownership controls	no	yes	no	yes	yes
4. Region controls	no	no	no	no	yes
5. Standard error of regression	0.101	0.097	0.102	0.098	0.097

Notes: Robust standard errors in parentheses. All models include a constant term. Dependent variable: log change in full-meal price. Prices include tax. Sample: 317 stores.

In the baseline specification with no controls, the New Jersey coefficient is 0.033 with a standard error of 0.014, indicating that full meal prices rose by approximately 3.3 percent more in New Jersey than in Pennsylvania. This suggests that roughly one-third of the wage cost increase was passed through to consumers in the form of higher output prices. The GAP specifications show similarly positive but less precisely estimated effects. Adding store and region controls modestly shifts the point estimates, but the qualitative pattern of partial price pass-through remains consistent across specifications.

4.6 Comparison with the Original Paper

Overall, our replication achieves a very high degree of concordance with the original results reported by Card and Krueger (1994). All core coefficient estimates match the published values to within one percent deviation. The cross-state DID estimate of 2.75 FTE is nearly identical to the original figure of 2.76. The baseline New Jersey regression coefficient of 2.33 matches the published value exactly, as does the GAP coefficient of 15.65. The estimated price effect of 0.033 is also identical to the original finding.

Minor discrepancies in third-decimal values and standard errors arise from slight differences in missing value treatment and the coding of temporary closures. These differences are negligible in magnitude and do not alter any substantive conclusions.

Taken together, the replication confirms that the original paper's core findings are highly robust and replicable under standard empirical protocols.

5. Independent Extension: Heterogeneity by Store Ownership

5.1 Research Question and Theoretical Framework

The baseline difference-in-differences estimator captures the average employment effect of the minimum wage across all stores, but this single average may mask important heterogeneity across organizational forms. Fast-food restaurants operate under two distinct ownership models—company-owned and franchised—that differ systematically in incentive structures, decision-making authority, and cost sensitivity. This extension tests whether the employment effect of the minimum wage differs systematically across these two types.

Theoretical predictions about adjustment patterns diverge based on organizational incentives. For franchised stores, franchisees are full residual claimants of store profits. A minimum wage increase directly compresses net income, creating strong incentives to improve labor efficiency through tighter scheduling, reduced idle time, and higher service throughput. These efficiency gains can offset higher wage costs and may even support expanded employment. Franchisees also retain greater autonomy over staffing levels and local pricing decisions, giving them more flexibility to adjust operations in response to cost shocks.

For company-owned stores, by contrast, store managers are salaried employees with only weak performance incentives. Standardized human resource policies and centralized pricing set by headquarters limit operational flexibility. As a result, employment responses to minimum wage hikes may be more muted in this group.

We test the hypothesis that the positive employment effect of the minimum wage is larger in magnitude for franchised stores than for company-owned stores.

5.2 Empirical Strategy

To examine heterogeneous employment effects by store ownership, we use a triple-difference (DDD) framework that augments the baseline DID design with a third dimension of variation: ownership type. The full specification includes all main effects and pairwise interactions of state, time, and ownership, plus the three-way interaction term that identifies the differential policy effect for company-owned relative to franchised stores. All specifications control for chain fixed effects to absorb systematic differences across restaurant chains, and use heteroskedasticity-robust standard errors.

We estimate the model on the same 357-store balanced sample used for the core regressions, to ensure direct comparability with the baseline results.

5.3 Results

Panel A of Table 8 reports mean employment and wage levels by state and ownership group for both survey waves. The raw difference-in-differences implied by these cell means is +3.27 FTE for franchised stores and +2.32 FTE for company-owned stores. The pattern is consistent with our theoretical hypothesis: franchised stores experience larger employment gains following the minimum wage hike.

Table 8: Heterogeneity Analysis by Store Ownership

Panel A. State and ownership descriptive statistics

Ownership	State	FTE wave 1	FTE wave 2	FTE diff.	Wage wave 1	Wage wave 2	Wage diff.
Franchised	Pennsylvania	26.005	23.603	-2.402	4.510	4.537	0.027
Franchised	New Jersey	20.825	21.693	0.867	4.598	5.081	0.483
Company-owned	Pennsylvania	18.652	16.385	-2.267	4.849	4.766	-0.082
Company-owned	New Jersey	19.709	19.764	0.054	4.639	5.080	0.442

Panel B. Triple-interaction FTE DID regression

Term	Coefficient	Robust SE	p-value
NJ (treat)	-4.084	1.979	0.039
Post	-2.244	2.266	0.322
Company-owned	-4.866	2.308	0.035
NJ × Post	2.642	2.408	0.273
NJ × Company-owned	4.708	2.551	0.065
NJ × Company-owned	0.328	2.835	0.908
NJ × Post × Company-owned	-0.888	3.221	0.783

Notes: Dependent variable: FTE employment. Model includes chain fixed effects. Robust standard errors are heteroskedasticity-robust. Sample: 357 stores × 2 waves = 714 store-wave observations.

Panel B presents the full triple-difference regression results. The baseline DID effect for franchised stores—captured by the NJ × Post interaction—is 2.64 FTE, similar in magnitude to the core replication estimate. The three-way interaction term (NJ × Post × Company-owned) is -0.888, indicating that the employment effect for company-owned stores is 0.89 workers smaller than for franchised stores. The sign of this point estimate aligns with our theoretical prediction that company-owned outlets exhibit weaker employment responses.

However, the triple interaction term has a p-value of 0.783 and is statistically indistinguishable from zero. We therefore cannot reject the null hypothesis that the employment effect of the minimum wage is identical across ownership types.

5.4 Discussion

The point estimates follow the expected theoretical pattern, but the difference in employment effects across ownership types is not statistically significant.

Three factors likely account for this. First, statistical power is limited: only 121 company-owned stores in the balanced sample reduce the ability to detect moderate effect-size differences. Second, the fast-food industry is highly operationally standardized, with similar job roles and staffing patterns across ownership types, leaving limited scope for divergent labor adjustments. Third, the 18.8 percent minimum wage increase, while meaningful, may not be large enough to trigger sharply different organizational responses.

Importantly, both groups show positive point estimates. The positive employment effect holds broadly across both ownership types, which strengthens the external validity of the core finding within the fast-food industry.

6. Validation, Limitations, and Conclusion

6.1 Internal Validity

Several lines of evidence support the causal interpretation of our employment results. First, the policy had a strong first-stage effect on wages: the minimum wage mandate sharply raised starting wages in New Jersey, with nearly all affected stores complying exactly with the $5.05 threshold. Second, a placebo test using only Pennsylvania stores yields no significant association between a hypothetical wage gap and employment changes, ruling out spurious correlation driven by common trends. Third, the positive employment estimate holds across a full battery of specification checks, including alternative variable definitions, sample restrictions, and estimation methods. Finally, within-state evidence corroborates the cross-state finding: stores in New Jersey facing larger required wage increases show larger employment gains, consistent with a causal effect rather than state-level confounding shocks.

6.2 Limitations

This study has five key limitations. First, industry specificity: the sample covers only fast food, a highly standardized low-wage sector, limiting generalizability to other industries. Second, short-run focus: outcomes are measured only 7–8 months post-policy; long-run effects may differ as firms adjust capital, technology, and entry-exit decisions. Third, narrow geographic scope: results are limited to New Jersey and eastern Pennsylvania, and may not hold in other labor market settings. Fourth, intensive margin emphasis: the analysis primarily captures employment changes among continuing stores; the small number of closures prevents precise

estimation of extensive margin effects. Finally, store-level data cannot capture within-store wage distributions or worker-level hours adjustments

6.3 Conclusion

This replication confirms the core finding of Card and Krueger (1994): the 1992 New Jersey minimum wage hike did not reduce fast-food employment. Employment rose modestly in New Jersey relative to eastern Pennsylvania, and the result holds across a wide range of specification checks. The wage increase was partially passed through to higher prices, with little evidence of adjustment along non-wage margins.

Our extension finds larger positive point estimates for franchised stores than for company-owned outlets, consistent with incentive-based theories of organizational response, but the difference is statistically insignificant. The core positive employment effect holds across both ownership types.

These findings challenge the standard neoclassical prediction that minimum wages necessarily reduce employment, and align with monopsonistic labor market or efficiency wage models where moderate minimum wage increases can boost productivity and employment.

References

Card, D., & Krueger, A. B. (1994). Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. American Economic Review, 84(4), 772–793.